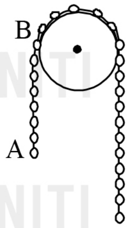
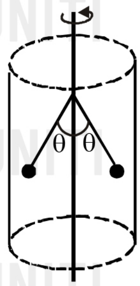
SUBJECTIVE TYPE

1. A homogeneous chain of length l hangs across the surface of a smooth peg such that fraction f of the chain hangs on the left side. If the chain is released from rest in the position shown in the figure, determine its speed v when end A passes through point B.
Neglect the radius of the peg.

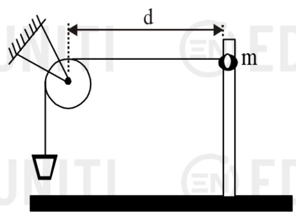


2. A governor consists of two 200 g masses attached by light, rigid 10 cm rods to vertical rotating axle (see fig). The rods are hinged so that the masses swing out from the axle as they rotate with it. However, when the angle θ is 45°, the masses encounter the wall of the cylinder in which the governor is rotating.

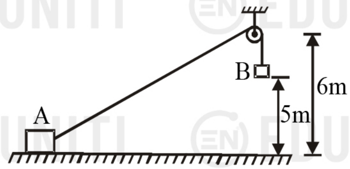


(a) What is the minimum rate of rotation, in revolutions per minute required for the masses to just touch the wall?
(b) If the coefficient of friction between the masses and the wall is 0.35, at what rate does the friction dissipate mechanical energy as a result of the masses rubbing against the wall when the mechanism is rotating at 300 rev/min?

3. A ring of mass $m = 3$ kg slides over a smooth vertical rod. Attached to the ring is a light string passing over a smooth fixed pulley at a distance of $d = 0.8$ m from the rod. At the other end of the string, there is a mass $M = 5$ kg. The ring is held level with the pulley and then released (see fig). Determine the distance by which the mass m moves down before coming to rest for the first time.

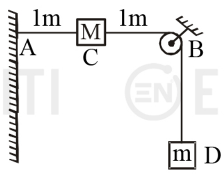


4. A block of mass m is held at rest on a smooth horizontal floor. A light frictionless, small pulley is fixed at a height of 6 m from the floor. A light inextensible string of length 16 m, connected at A passes over the pulley and another identical block B is hung from the string. Initial height of B is 5 m from the floor as shown in fig. When the system is released from rest, B starts to move vertically downwards and A slides on the floor towards right. (Take $g = 10\ \text{ms}^{-2}$)

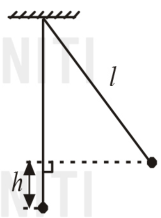


(a) If at an instant string makes an angle θ with horizontal, calculate relation between velocity u of A and v of B.
(b) Calculate v when B strikes the floor.

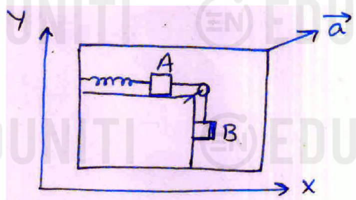
5. A string with one end fixed on a rigid wall, passing over a fixed frictionless pulley at a distance of 2 m from the wall, has a point mass M of 2 kg attached to it at a distance of 1 m from the wall. A mass m of 0.5 kg is attached to the free end. The system is initially held at rest so that the string is horizontal between wall and pulley and vertical beyond the pulley as shown in fig. What will be the speed with which the point mass M will hit the wall when the system is released? (Take $g = 10\ \text{ms}^{-2}$).



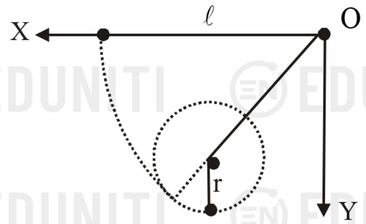
6. A small positively charged ball of mass m is suspended by an insulating thread of negligible mass. Another positively charged small ball is moved very slowly from a large distance until it is in the original position of the first ball. As a result, the first ball rises by h. How much work has been done on the system?



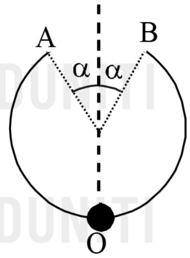
7. The arrangement shown in fig. is at rest. An ideal spring having spring constant k = 2 Nm^{-1} is connected to block A. Blocks A and B are connected by an ideal string passing through a frictionless pulley. The mass of each block A and B is m = 2 kg. When the spring was in natural length, the whole system is given an acceleration $\vec{a} = (4\hat{i} + 3\hat{j})\ \text{ms}^{-2}$ as shown. If coefficient of friction of both surfaces is $\mu = 0.25$, then find the maximum extension (in metres) of the spring. (Take g = 10 ms^{-2})



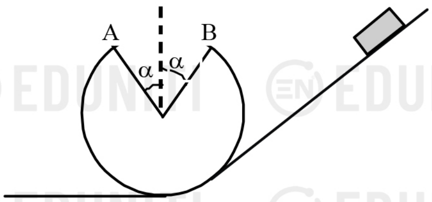
8. A simple pendulum of length l is suspended by a nail on a vertical wall at point O. Another nail is to be fixed on the wall such that if the pendulum is released from its initial horizontal position, it just completes a vertical circle round that nail as shown. Find the locus of the points where the second nail can be fixed.



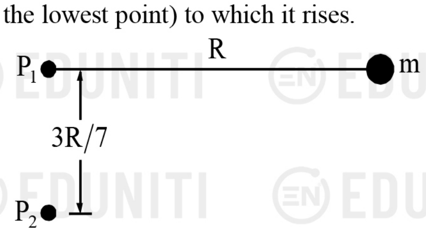
9. A wire is bent along an arc with a radius R (see fig.). A bead is placed on the wire that can move along it without friction. At the initial moment, the bead was at point O. What horizontal velocity should be imparted to the bead for it to get onto the wire again at point B after flying a certain distance AB through the air.



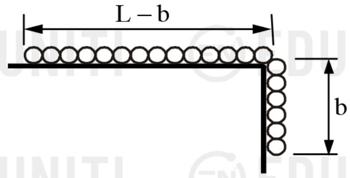
10. A small body slides down an inclined surface passing onto a loop (see fig.) from the minimum height ensuring that the body does not leave the surface of the loop. What symmetrical segment with an angle $\alpha < 90^o$ can be cut out of the loop for the body to reach point B after traveling a certain distance in the air?



11. A pendulum consists of a bob of mass m and a string of length R suspended from a peg P_1 on the wall. A second peg P_2 is fixed vertically below the first one at a distance of $3R/7$ from it (see fig). The pendulum is drawn aside such that the string is horizontal, and released. Calculate the highest point (with respect to the lowest point) to which it rises.



12. The chain shown in the fig. starts from rest with a sufficient number of links hanging over the edge to barely initiate the motion in overcoming friction between the remainder of the chain and the horizontal supporting surface. Determine the velocity v of the chain as the last link leaves the edge. The coefficient of kinetic friction is μ_k. Neglect any friction at the edge.

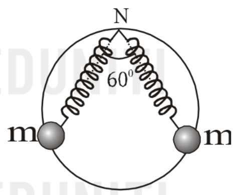


PARAGRAPH TYPE

Paragraph for Question Nos. 13 to 15

Two identical beads are attached to free ends of two identical springs of spring constant $k = \dfrac{(2+\sqrt{3})mg}{\sqrt{3}\ \ R}$.

Initially both springs make an angle of 60° at the fixed point N. Normal length of each spring is 2R. Where R is the radius of smooth ring over which bead is sliding. Ring is placed on vertical plane and beads are at symmetry with respect to vertical line as diameter.



13. Normal reaction on one of the bead at initial moment due to ring is

(a) mg/2 (b) $\sqrt{3}$ mg/2

(c) mg (d) Insufficient data

14. Relative acceleration between two beads at the initial moment

(a) g/2 vertically away from each other

(b) g/2 horizontally towards each other

(c) $2g/\sqrt{3}$ vertically away from each other

(d) $2g/\sqrt{3}$ horizontally towards each other

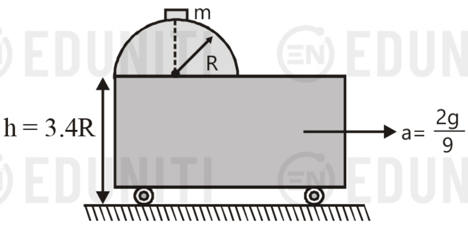
15. The speed of bead when spring is at normal length is

(a) $\sqrt{(2+\sqrt{3})gR/\sqrt{3}}$ (b) $\sqrt{(2-\sqrt{3})gR/\sqrt{3}}$

(c) $\sqrt{2gR/\sqrt{3}}$ (d) $\sqrt{3gR}$

Paragraph for Question Nos. 16 to 18

A vertical frictionless semicircular track of radius R is fixed on the edge of movable trolley. Initially the system is at rest and a mass m is kept at the top of the track. The trolley starts moving to the right with a uniform horizontal acceleration a = 2g / 9. The mass slides down the track, eventually losing contact with it and dropping to the floor h below the trolley.



16. The angle θ with vertical, at which it losses contact with the trolley is

(a) 37° (b) 53°

(c) $\cos^{-1}\left(\frac{2}{3}\right)$ (d) $\frac{\pi}{2}-\cos^{-1}\left(\frac{2}{3}\right)$

17. The speed of mass with respect to trolley when it crosses contact with it is

(a) $\sqrt{gR}$ (b) $\sqrt{2gR}$

(c) $\sqrt{2gR/3}$ (d) $\sqrt{gR/3}$

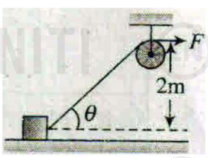
18. The time taken by the mass to drop on the floor, after losing contact is

(a) $\sqrt{2R/3g}$ (b) $\sqrt{6R/g}$

(c) $\sqrt{3R/2g}$ (d) $2\sqrt{3R/g}$

Paragraph for Question Nos. 19 to 21

A force F = 50 N is applied at one end of a string, the other end of which is tied to a block of mass 10 kg. The block is free to move on a frictionless horizontal surface. Take initial instant as $\theta = 30^o$ and final instant as $\theta = 37^o$. For the time between these two instants, answer the following questions



19. Net work done by the force F on the block is

(a) 50 / 3 J **(b)** 100 / 3 J

(c) 75 J (d) none of these

20. What is the final velocity of the block if initially it was at rest

(a) $\sqrt{25/3}\, ms^{-1}$ **(b)** $\sqrt{20/3}\ ms^{-1}$

(c) $5ms^{-1}$ (d) none of these

21. Find the ratio of initial acceleration to final acceleration of the block.

(a) $3\sqrt{5}/8$ (b) $8\sqrt{3}/5$

(c) $8\sqrt{5}/3$ (d) $5\sqrt{3}/8$

MORE THAN ONE CORRECT

22. A particle moves in one dimensional field with total mechanical energy E. If potential energy of particle is U(x), then

(a) particle has zero speed where U(x) = E

(b) particle has zero acceleration where U(x) = E

(c) particle has zero velocity where $\frac{dU(x)}{dx} = 0$

(d) particle has zero acceleration where $\frac{dU(x)}{dx} = 0$

23 A particle of mass 5 *kg* moving in the *X-Y* plane has its potential energy given by $U = (-7x + 24y)$ Joule. The particle is initially at origin and has velocity $\vec{u} = (14.4\hat{i} + 4.2\hat{j})$ m/s. Then,

(a) the particle has speed 25 m/s at t = 4 sec

(b) the particle has an acceleration 5 m / s^2

(c) the acceleration of particle is normal to its initial velocity

(d) none of these

24 The acceleration of a particle with speed v is $\vec{a} = \vec{b} \times \vec{v}$, where b is a constant vector which is neither parallel nor anti-parallel to velocity. Then,

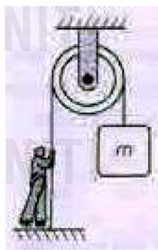
(a) the velocity of the particle remains constant

(b) the work done on the particle is zero

(c) the speed of the particle remains constant

(d) the work done on the particle is non-zero

25 A man is slowly pulling a string connected to a block of mass m as shown in the figure. During an upward motion of the block,



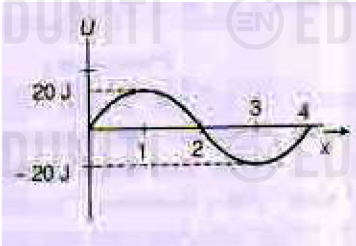
(a) the work done by the string on the block is positive

(b) the work done by the man on the string is negative

(c) the work done by string on the man is zero

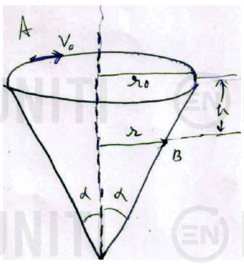
(d) the net work done on the block must be positive

26 A particle having potential energy shown in graph is moving towards right. The initial kinetic energy of particle is 21 J at x = 0. If only conservative forces are acting on the particle, mark the correct option(s)



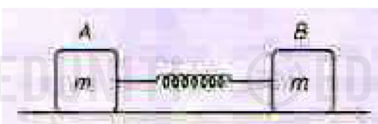
(a) The particle accelerates for $0 \le x < 1\text{m}$
(b) At x = 1 m, the particle has minimum kinetic energy
(c) The kinetic energy is 41 J at x = 3 m
(d) The particle is trapped within $0 < x \le 4$

27 A small particle of mass m is given an initial velocity v_0 along tangent to the horizontal rim of a smooth cone of radius r_0 from the vertical centre line as shown in figure. As, the particle slides to point B, a vertical distance h below A and distance r from vertical centre line, its velocity v makes an angle θ with the horizontal tangent to the cone through the point B.



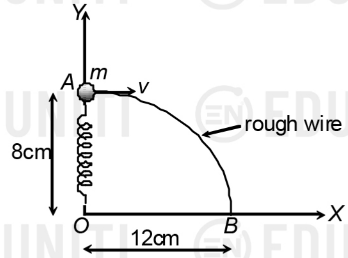
(a) The speed of particle at point B is $\sqrt{v_0^2 + 2gh}$
(b) The value of v_0 for which particle will be moving in a horizontal circle at point A is $\sqrt{gr_0 \cot\alpha}$
(c) If the particle moves in a horizontal circle at point B, then $r = (v_0^2/g + 2h)\tan\alpha$
(d) Work done by normal reaction on the particle is zero

28 Two identical blocks each of mass m are connected by a light spring of constant k = 100 N/m. Initially, spring is compressed upto 0.2 m and then released. Which of the following statements is / are correct



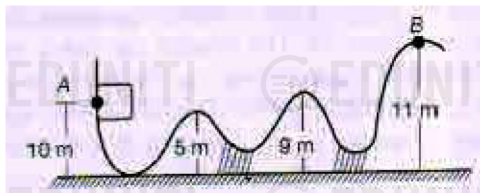
(a) The work done by the spring on the system till the spring comes in natural length is 1 J
(b) The work done by the spring on the system till spring comes in natural length is 2 J
(c) The work done by the spring on the system till the spring gets stretched to the maximum
(d) The work done by the spring on the system is zero till the spring gets stretched to the maximum is – 2 J

29 A ring of mass 200 gram is attached to one end of a light spring of force constant 200 N/m. The ring is constrained to move on a rough wire in the shape of quarter ellipse of major axis 24 cm and minor axis 16 cm with its centre at origin. The plane of ellipse is vertical and wire is fixed at points *A* and *B* as shown in figure. Initially ring is at *A* with other end of spring fixed at origin. If normal reaction of wire on ring at *A* is zero and ring is given a horizontal velocity of 2 m/s towards right so that it just reaches point *B*, then select the correct alternative(s) ($g = 10$ m/s²)



(a) Natural length of the spring is 9 cm
(b) Work done by gravity is 0.16 J
(c) Work done by spring is -0.08 J
(d) Work done by friction is -0.48 J.

30 A small block is released from rest from a height of 10 m on a smooth ramp. The block does not fly off in air (g = 10 m/s²). Friction is absent everywhere.



(a) The maximum speed of the block during motion is $10\sqrt{2}\ \text{m/s}$
(b) The block repeats its path continuously
(c) Total mechaincal energy of the block remains constant
(d) The block will never reach at the point B